Exam 1 Study Guide

Research Methods in Psychology

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1 Introduction

This is the instructor-provided study guide for the midterm exam for PSY-300. Please use this document as **one** of the aids in preparing for this exam, but I strongly encourage you to develop additional study aids and tools on your own. This document is **not meant to be comprehensive** to all of the content covered in modules 1 through 6 - it is a merely a summary of what was covered, and meant as a review tool to help guide your practice.

Midterm / Exam 1 Structure covers the basic policies and procedures of the test, with information lifted directly from the syllabus. Tips for Preparing highlights what I feel are good practices to employ as you get ready to take the test. Tips for Studying contains ideas on what additional steps you can take to help get ready for the exam and [Responsible Uses of AI Tools for Studying] explains some of the ways you may responsibly use AI to help you prepare for the exam. Practice Questions are provided to help you recall and practice some of the major concepts from the modules. I make no guarantee that the content of the Practice Questions will be covered on the test, but believe they help give a good general idea of the different concepts we covered throughout the modules.

Please do feel free to reach out with any questions you may have, but also make ready use of these many opportunities and resources to do well. I believe in each of you to use your time and diligence to turn in an excellent performance on this test.

2 Midterm / Exam 1 Structure

Please review the following information and procedures regarding the midterm and final exams in this course. The following text is also in the syllabus.

There may be additional section-specific policies relevant to the exam; please check the syllabus for the most accurate description of the exam as it is relevant to your section and reach out to the instructor via email with any additional questions

There will be 2 exams in this course, effectively a midterm and a final. These exams are intended to be **cumulative** and will cover content from all units and quizzes. In practice, this means that the midterm will only cover the first half of the course, and the final will only cover the content for the entire course.

The format is as follows:

- Each exam is 50 multiple-choice questions, 2 points for each question
- Exams will contain content from the entire unit, between all lectures AND readings and any other activities
- Exams are timed, 60 minutes only (used to be 50 minutes)
- Exams are not open-note, open-book, or collaborative. You are not permitted to use any form of assistance other than what is explicitly provided by the instructor. Do not discuss the tests with other students after it has concluded

- You should not be speaking, texting, or calling with other people during the exam. You should not be holding, using, or referring to any phone, tablet, or other electronic device during the exam. You should not be wearing any headphones, earbuds, headsets, or other ear devices (unless it is pre-approved as an accommodation for medical reasons)
- Failure to adhere to the above guidelines may result in an automatic failure on the exam, if it raises the suspicion of un-authorized help during the exam, at the discretion of the instructor. Multiple failures to adhere to the rules of the exam may result in dismissal from class and referral to GVSU administration for further disciplinary action.
- Exams will be graded promptly and reviewed as soon as possible; the correct answers to the questions will also be provided.

3 Tips for Preparing

The below pieces of advice are my opinionated thoughts about how you may successfully prepare for this exam. Each student may find different strategies and techniques useful, please customize the following advice to your own needs.

- Because the exam is fairly comprehensive, you should make sure to review any materials relevant to the content we've covered in class, which may include:
  - Any and all lecture notes taken, whether that be the guided notes filled out or your own personal notes
  - Printed copies of your lecture check-ins
  - Any notes or activities you've completed from the book as additional practice
  - Any additional handwritten or student-typed preparation, such as answers to this study guide
- If you suspect that you may be absent or miss the exam window, you should contact the instructor as soon as possible
- If you have a documented accommodation for this course, you should work with the instructor to ensure that the exam format will be adequately address your needs, and work to make other arrangements for taking the test if necessary
- Try to get a good night of sleep for the several nights leading up to the exam
- Follow your usual diet leading up to the exam; try not to eat new foods that may upset your stomach and/or disrupt your attention while you attempt to exam
- If you normally consume it, keep your usual habits of consuming a healthy dose of caffeine leading up to the exam
- Dress in layers that you can add or take off to remain comfortable temperature-wise, during testing
  - However, do not wear any hoods up or hats that obscure your ears and may raise concern that you are wearing any earbuds

The below bullet points are only relevant to online sections of this course:

- You should take the exam on a laptop or desktop computer, not a tablet or phone. You should have your computed plugged in during the exam to reduce risk of battery running out, and should warn others you live with to not disturb you until you are done
- Find a quiet, secluded spot with good internet connection and an outlet for your computer. You may want to take the exam in a different place than your dorm room, if you feel that your roommates may not give you adequate private space to take the exam
- Place any other devices and distractions (e.g., phone, earbuds, etc.) in another room or away in a bag, unless necessary for medical or academic accommodations
- Please do not have any snacks or drinks while taking the exam, as their packaging may raise the concerns of cheating. Keep your workspace entirely clear of any materials or devices that could be perceived as cheating

4 Tips for Studying

The below pieces of advice are my opinionated thoughts about how you may successfully study for this exam. Each student may find different strategies and techniques useful, please customize the following advice to your own needs.

- As a general rule: **create, don't consume!** The best study materials are the ones that you make and develop. While I give you this document as a starting point, you should make the notes and study materials that are best suited to *you*. The process of developing study materials on your own will help cement the content for you.
- **Give yourself enough time to study!** Good studying starts early, and remains consistent. There is simply too much material to making cramming an effective strategy. Starting studying only the night before an exam will likely be insufficient for getting you a good grade.
- **Be honest with yourself about your ability on the content!** Too many students seem to "gaslight" themselves into believing they "know" the content, just by re-reading their notes (I've done it myself). It never hurts to test or quiz yourself and admit when you struggle on a certain area or topic.
- **DO NOT use this study guide as your sole resource when studying.** I provided this guide as a convenience, but I make no certain guarantee that it contains any and all information relevant to the exam. Any of the content covered in lectures or reading is fair game to be tested upon - and you should use resources from those things to prepare. Completion of this study guide does not necessarily guarantee success on the actual exam.
- Try "teaching" the materials to an (un)willing volunteer or inanimate object

  - Trying to teach through and explain a concept will immediately make it apparent whether you understand it, or do not. This is how I double-check myself when writing lectures: if I can't explain something fluidly, then I haven't truly mastered explaining it well enough yet.
  - You could even try using my slides and see if you can talk "between the lines" and go beyond what is just written down - lecturing on a certain piece of content can actually really help you reinforce your understanding.

- Review past quizzes and lecture check-in answers; ensure you have corrected your understanding where you may have gotten a question wrong the first time through

- Consider "re-annotating" your previous notes:
  - Add question marks where you are confused on something
  - Add highlights to parts that stick out as especially critical to understanding
  - Re-work discussion and multiple choice questions strewn throughout the lecture guided notes and make sure you have good answers for each of them

- Complete practice and review questions at the end of each chapter of the textbook, your textbook has many useful activities for thinking through and practicing the concepts of the class; many students do not realize how useful these practices can be for preparing

- Consider coming to office hours the instructor to clarify difficult topics

5 Responsible Uses of AI Tools in Studying

I generally caution against use of AI in studying and the educational process in general, because many students (in my experience) use it in a way that deprives them of learning, rather than enhancing it. *But,* I do believe there is a way to use it to assist in the learning and memorization process. Let me give some examples of what to do and what not to do:

- As a general rule, **always double-check with your notes and textbook to ensure you get accurate answers - AI IS NOT INFALLIBLE**

- Avoid using plain 'explanation' prompts with no follow-up:
  - E.g., "Explain behaviorism to me", "Tell me the lobes of the brain"
  - While such a question may *feel* like it helps, you likely won't actually remember what it says unless you use that explanation to do something like write a definition in your own words
  - This is a good starting point for clarifying things, but won't help in the broad scheme of things past that

- Use it to generate tools like mnemonics or tricks for assisting memorization:
  - E.g., "Make me an easy-to-remember acronym or mnemonic for remembering the lobes of the brain and their general functions"

  - If you actually practice memorization with these shortcuts, you can likely really benefit from these!

- Use it to make some mock or test questions based on your notes:
  - E.g., Upload your notes and ask, "Make me some example multiple-choice and short answer questions based on this document, and give me explanations for each one"
  - May have mixed results depending on how well the AI can "read" your notes
  - Make sure you actually test yourself and correct your understanding, not just using the answer key to make you "think" you know the correct answer, when you can't really explain it

- Use it to make short passage examples of certain principles in practice:
  - E.g., "Write me 4 examples of classical conditioning of behavior using positive reinforcement, negative reinforcement, positive punishment, and negative punishment"

6 Practice Questions

6.1 Week 1

- What is a research **producer**? What is a job title this person is likely to have? What sorts of activities is this person likely to perform?
- What is a research **consumer**? What is a job title this person is likely to have? What sorts of activities is this person likely to perform?
- Describe the procedures, findings, and structure of Harlow's monkey study to investigate cupboard theory vs. contact comfort theory.
- Describe the concept of **Empiricism** and its relationship to the **Theory-Hypothesis-Data Cycle.** Define and explain each individual part of the cycle.
- How are Empiricism, the process of **reproducing** scientific studies, and the **self-correcting** nature of science related?
- What are **Merton's 4 Scientific Norms**? Be able to give definitions of each and examples of behaviors that support these norms.
- Why don't we use the word "prove" in science, and why do we prefer using the **weight of evidence** to describe support for a claim? Write out an example of what an improper claim statement (i.e., one being too certain) may look like, and then give a "fixed" version of that statement.
- What are the 3 contexts/types of research? Give an example of a study that would be described as each type.

- Compare and contrast the methods and trust-worthiness of **scientific journalism** and **scientific articles published in peer-reviewed journals.** See if you can find an example of both types, or better, find a piece of journalism and then the original article the journalism was based upon.
- What citation style do we use in this course? What are the core, important components of a full citation (Like that found in a "References" section)? Looking at the information of an article, try to type out the correct formatted citation for it. What are some shortcuts in Google Scholar or scientific databases to get the citation information?

6.2 Week 2

- What are the 4 sources of knowledge or information we can draw from? Which of these is the most sound?
- Why do we consider research to be superior to **authority**? Provide 2 examples of faults in trusting authority.
- Why do we consider research to be superior to **personal experience**? Provide 2 examples of faults in relying on personal experience.
- Why do we consider research to be superior to **intuition**? Provide 2 examples of biases in intuition.
- Describe the dangers of **confounds** and having no **comparison groups.** What issues do these introduce? Describe an example of a confound.
- What does is mean that research is **probabilistic**? How does this affect how we apply research findings to individual people?
- Describe the 6 biases we discussed in intuition. Write an example of each bias.
- What is the difference between an **original empirical article,** a **literature review**, and a **meta-analysis?** Try to find an example of each of these using the techniques we learned to find research.
- What type of search engine should I use in order to find *peer-reviewed* journal articles? Give two examples of appropriate tools. Find a scientific article that is specifically about traumatic brain injury and recovery.
- What are the procedures for fully inspecting a research article before using it?
- Why are books in science generally less valued as a source, compared to scientific journal articles? Connect this back to criticisms of authority.

6.3 Week 3

- Define a **variable** and a **constant**. Give an example of each in a research setting.

- What are the 3 different **scales of measurement** for variables? Give an example of a variable of each type.
- What is the difference between a **measured** and **manipulated** variables? What are some reasons a measure may have to measured, and could not be manipulated?
- Compare and contrast **construct** and **operational** variables. What are examples of both of these? For the construct of anxiety, propose a simple operational measure.
- What are the 3 types of **claims** we can make? Give one example of each type and a few keywords that signal each type of claim. Which of these claims requires the "hardest" burden to investigate?
- What statistical and graphical methods do we have for presenting information about associative claims? How does one interpret each of these?
- What are the three specified criteria that must be met that allow us to determine causation between two or more things?
- What are the 4 **claim validities** for examining scientific claims? For each validity, give one example of a detriment and one example of a benefit. Which of the validities is only really applicable to causal claims?

6.4 Week 4

- Be able to explain the general structure and goals of the **Tuskegee Syphilis Study** and the **Milgram Obedience Studies**.
- Describe the questionable/bad ethics of the Tuskegee Syphilis Study and the Milgram Obedience Studies. In what way did these studies cause undue harm or fail the participants? Connect these failing to specific principles or guidelines in the **APA Code of Ethics**.
- Enumerate the 3 principles of the **Belmont Report** and the 5 principles of the APA Code of Ethics. Be able to define each of these *in detail.*
- Describe the relationship between **informed consent**, **deception**, and **debriefing**. In what cases would deception be allowed?
- What are the governing bodies for ensuring ethical research procedures at any institution? In what ways can they influence and regulate ongoing projects?
- The first chapter mentioned **"Pre-registration"** of hypotheses - which of the APA Code of Ethics does this best relate to?
- Describe a situation where ethics and study validity may be at odds or in conflict. What is an example of a study that would be scientifically valid, but not ethical?

6.5 Week 5

- What type of variable requires a **conceptual definition?**
- What are the 3 types or mediums of psychological measures? Provide an example of each type and describe what these might look like in practice.
- What is the relationship between measurement **reliability** and **validity**? What claim validity do these both fall under?
- Describe the concept of reliability: what is a good synonym for it? Provide a real life example of something "unreliable" in the measurement sense. What are the 3 types of reliability discussed in class?
- What is the most common graphical and statistical method to assess reliability? How does one interpret each of these graphs and statistics? (You do not need to know how to calculate statistics, but you should know what certain values would imply)
- Define measurement validity - what is a good synonym for it? What are the 5 types of measurement validity we discussed in class? Which of these are the "weakest" or the most subjective?
- What are the two methods by which to investigate **criterion validity**? What sorts of analyses do we use in the case of either type?
- Using the visual analogy of targets (as presented in class), what would a reliable, but invalid tool look like?
- By what methods can we find established measures for common psychological constructs? How is this similar to searching for scientific literature?

6.6 Week 6

- When we use the terms **survey** or **pool** what type/medium are we referring to? What are the roughly 4 types of questions we talked about in class? Write an example of each question type.
- What are the 4 threats we discussed in how questions are written? What are the available solutions to address each one of these threats?
- Describe what a **response set** is and why it is undesirable in participant responses. What are the 3 to 4 response sets we talked about, and how can we attempt to prevent them in our research?
- Describe the concerns in long-term memory and peoples' ability to introspect on *why* they do something, which could affect our survey results?
- Describe the difference between an **observational measure** and a **self-report** one. Why might a psychologist see one as more "objective" than the other?

- What are the 3 major concerns to measurement validity in observational measures? Give an example of each one of these occurring and how we might solve each.
- In what ways may we choose to use deception in observation tools to enhance validity?